

Marshal Turner

Machine Learning Engineer | Computer Vision & Production ML
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Machine Learning Engineer with 3.5+ years of experience designing, training, and deploying production ML models that drive measurable business impact. Expertise in end-to-end ML pipelines, from data preprocessing and feature engineering to model training, evaluation, and scalable deployment using PyTorch and TensorFlow. Hands-on computer vision experience developing image classification systems on large-scale real-world image datasets, including robustness testing under challenging conditions. Proven collaborator with cross-functional teams, delivering automated pipelines and clear technical guidance to stakeholders. Pursuing an MS in Analytics at Georgia Tech.

Technical Skills

Languages & Frameworks: Python, SQL, R, PyTorch, scikit-learn, LangChain, TensorFlow/Keras, LangChain

Statistics & Experimentation: A/B Testing, Regression, Observational Causal Inference, Propensity Modeling, Metrics Development & Validation, Experiment Design

ML & Modeling: Classification, Time-Series Forecasting (ARIMA), Predictive Modeling, Catboost/XGBoost

MLOps & Cloud: Docker, AWS, Azure, Snowflake, GitHub, CI/CD pipelines, API Services, Inference Pipelines

Data & Visualization: Pandas, Matplotlib/Seaborn, Tableau, Power BI, Selenium, Django

LLMs & RAG: Retrieval-Augmented Generation (RAG), Prompt Engineering, Groq LLMs, HuggingFace Transformers, ChromaDB, Embedding Models, LLM Evaluation (faithfulness & relevancy)

Computer Vision: Image Classification, Image Preprocessing, Pixel Feature Engineering, PCA/ISOMAP

Professional Experience

Flexential Inc. - Remote

Data Scientist (BI Developer), July 2024-April 2026

- Designed, built, and operated a production HistGradientBoosting classifier for product profitability using revenue, cost, and location data; achieved 91% precision / 89% recall and directly informed product and pricing decisions.
- Developed and deployed a production Propensity Lead Classifier using causal-style inference and propensity modeling; generated >\$4M annual revenue and outperformed two third-party vendors on F1-score while influencing cross-functional strategy.
- Mentored intern on end-to-end NLP sentiment classifier; built automated inference pipeline that reduced manual review effort by 50%+ across ~5,000 tickets/month and improved product decision-making.
- Led design and implementation of API integration pipeline for 22 MW contracted capacity tracking; delivered production-grade inference and forecasting tooling that enabled accurate strategic planning.
- Co-authored company Data Science Charter with SVP; served as strategic thought partner to automate deployment pipelines and streamline cross-functional project tracking.

Nolan Transportation Group, Inc. (NTG) - Hybrid

Data Science Analyst, November 2022 - April 2024

- Inherited, productionized, and operated an outsourced lane-reliability classifier; translated statistical insights into product decisions that optimized revenue on >60% of bids.
- Built and maintained ARIMA time-series forecasting model; achieved 1.9% MAPE and informed future experimentation and classifier logic.
- Designed and executed A/B test on calculated pricing field; reduced forecasting MAPE by 1.6% through rigorous statistical validation and production rollout.
- Developed Django-based internal application with multi-source ETL pipeline; created and validated C-level metrics that drove data-informed decisions for \$139M revenue portfolio.

Pricing BI Analyst, 2021 - November 2022

- Automated JIRA task creation via REST API + Outlook parsing; delivered production application that saved ~\$50K/year in labor.
- Built and maintained SSRS/Tableau reports and Excel macros for pricing automation and geographic lane analysis.

Projects

AWS DeepRacer PPO Agent - Georgia Institute of Technology, OMSA - DEC 2025

- Designed and trained a stabilized PPO policy across multiple scenarios; conducted 75k+ episode experiments and ablation studies while translating custom mathematical reward functions into production-grade code.
- Containerized entire training and inference pipeline in Docker; enabled reproducible experimentation and policy transfer across environments.

Overcooked Environment with MARL QMIX Architecture - Georgia Institute of Technology, OMSA - DEC 2025

- Designed and trained two cooperative QMIX agents; engineered reward shaping and conducted ablation studies that demonstrated emergent coordination behaviors through rigorous experimentation.

Real-World Fruit Classification (Computer Vision) – Georgia Institute of Technology, OMSA - OCT 2024

- Built and optimized image classification models on a 225k+ image Fruits-262 dataset (7.8 GB); performed extensive preprocessing, resizing (208x256 → 13x16), normalization, and dimensionality reduction via PCA + ISOMAP on pixel features.
- Trained and hyper-tuned multiple classifiers (Random Forest, KNN, SVM OVR, CART) on high-dimensional image data; selected best model based on precision, recall, and F1-score.
- Conducted real-world robustness testing by photographing fruits in a grocery store (Kroger) under challenging conditions (varying lighting, occlusions, backgrounds); achieved 53% accuracy on out-of-distribution images.
- Demonstrated end-to-end CV pipeline skills: data acquisition, feature engineering, model evaluation, and production-like robustness validation.

CR7 Fan LLM - Retrieval-Augmented Generation (RAG) LLM System - MAR 2026

- Built a full RAG pipeline using Chroma + HuggingFace MiniLM embeddings to store and retrieve 120 curated facts with low-latency, persisted vectorstore loading
- Designed a persona-driven Groq LLM with guardrails to prevent hallucinations and enforce factual grounding
- Tuned prompts, temperature, and retrieval parameters to balance creativity with factual accuracy
- Evaluated model outputs using DeepEval faithfulness and answer relevancy metrics, with a second Groq model acting as a neutral judge
- Deployed an interactive Streamlit app and resolved real-world engineering issues including dependency conflicts, model overhead, latency, and vectorstore persistence

Education

Georgia Institute of Technology: Master of Science in Analytics

Kennesaw State University: BS in Systems and Industrial Engineering

Expected May 2026

May 2018